Kushal Kafle

Senior Research Scientist, Adobe Research

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Research Interests

Multimodal LLMs / vision-language models (VLMs), representation learning, foundation model training on web-scale data, benchmarks, evaluation and model probing, robustness and fairness

Experience

Adobe Research, San Jose, CA *Mar 2020 – present*

Senior Research Scientist 2 (2026); Senior Research Scientist (2024); Research Scientist 2 (2022); Research Scientist (2020)

- Lead Adobe's vision-language foundation models across the full lifecycle (data, legal, training infrastructure, algorithms, and tech transfer): image-text embedding models and, recently, multimodal LLMs / VLMs.
- Our embedding model matches or beats the best open-weight models (e.g., Perception Encoder, SigLIP 2) on text-to-image retrieval benchmarks, by a wider margin on Adobe content.
- Mentored 28 research interns (2020–2026), mostly PhD students (full list).

Research Intern, Microsoft Research (2019); Adobe Research (2017 – 2018)

Education

Ph.D. in Imaging Science, Rochester Institute of Technology *2014 – 2020*

Advisor: Christopher Kanan. Thesis: *Advancing Multi-modal Deep Learning: Towards Language-grounded Visual Understanding*

Selected Technology Transfers

- **Adobe Express and Adobe Stock** *(2023 – present)*: Natural-language and image search for Express templates (+46% click-through) and Stock assets; "More like this" template similarity.
- **Adobe Lightroom** *(2025 – present)*: Natural-language photo search, plus assisted culling and Auto Stack via a compact on-device model.
- **Adobe Firefly Image Generation** *(2023 – 2025)*: Reference features in Firefly Image 1–3 (Style Reference, similar images); NSFW safety filter for image and video; training-data filtering.
- **Premiere Pro Media Intelligence** *(2025)*: Visual search in the Media Intelligence and Search panel.
- **Other:** Firefly image-to-video (prompt coherence and safety); AEM Smart Tags; Photoshop prompt-free object compositing.

Selected Publications

32 peer-reviewed papers; selected below. Full list on Google Scholar and kushalkafle.com.

[1] Fioresi, J., Heilbron, F. C., Nathani, P., Shah, M., & **Kafle, K.** (2026). Controlling Embedding Spaces with Text-Conditioned Transformations. *ECCV 2026*.

[2] Hernandez, J., Shi, J., Jenni, S., Ordonez, V., & **Kafle, K.** (2025). Improving Large Vision and Language Models by Learning from a Panel of Peers. *ICCV 2025*.

[3] Slyman, E., Lee, S., Cohen, S., & **Kafle, K.** (2024). FairDeDup: Detecting and Mitigating Vision-Language Fairness Disparities in Semantic Dataset Deduplication. *CVPR 2024*.

[4] Sameni, S., **Kafle, K.**, Tan, H., & Jenni, S. (2024). Building Vision-Language Models on Solid Foundations with Masked Distillation. *CVPR 2024*.

[5] Tanjim, M. M., Singh, K. K., **Kafle, K.**, Sinha, R., & Cottrell, G. W. (2024). Discovering and Mitigating Biases in CLIP-Based Image Editing. *WACV 2024*. Oral Presentation (2.6% of submissions)

[6] Shrestha, R., **Kafle, K.**, & Kanan, C. (2022). OccamNets: Mitigating Dataset Bias by Favoring Simpler Hypotheses. *ECCV 2022*. Oral Presentation (2.7% of submissions)

[7] Hayes, T. L.*, **Kafle, K.***, Shrestha, R.*, Acharya, M., & Kanan, C. (2020). REMIND Your Neural Network to Prevent Catastrophic Forgetting. *ECCV 2020*. *equal contribution

[8] **Kafle, K.**, Price, B., Cohen, S., & Kanan, C. (2018). DVQA: Understanding Data Visualizations via Question Answering. *CVPR 2018*.

[9] **Kafle, K.** & Kanan, C. (2017). An Analysis of Visual Question Answering Algorithms. *ICCV 2017*.

Professional Service

- **Area Chair**: ICCV 2025 (Lead AC), CVPR 2026 (Lead AC), CVPR 2027, WACV 2024–2026. **Senior PC**: AAAI 2027. **Workshops organized**: SiVL (ECCV 2018, NAACL 2019), DIRA (ECCV 2022).
- **Reviewer**: 100+ papers (CVPR, NeurIPS, ICCV, ECCV, ICML, ICLR, ACL). Outstanding Reviewer: CVPR 2018–21, NeurIPS 2021, ECCV 2026.

Updated Nov 2026